

LESSON 5-3 • SESSION 2 • PRACTICE • FORM A

The Trapezoid Area Formula

Learning target – I can find the area of a trapezoid using the trapezoid area formula. • 6.G.A.1

REMEMBER

- $A = \frac{1}{2} \times (b_1 + b_2) \times h$. The bases b_1 and b_2 are the two **parallel** sides.
- **Add** the bases first, multiply by the height, then take **half**. Slanted sides are not used.
- Missing base: double the area, divide by the height, then subtract the base you know.

WORKED EXAMPLE

Bases 6 m and 12 m, height 4 m.

Step 1 Add the bases: $6 + 12 = 18$.**Step 2** Multiply by the height: $18 \times 4 = 72$.**Step 3** Take half: $72 \div 2 = 36$.**Answer** $A = 36 \text{ m}^2$ **WORDS** base (base) • formula (fórmula) • right trapezoid (trapezio rectángulo)

PRACTICE

show your work • label units

1. Find the area of each trapezoid with bases
- b_1
- and
- b_2
- and height
- h
- .

a. 9 cm, 15 cm; $h = 8 \text{ cm}$ ____ cm^2

b. 20 in, 28 in; $h = 15 \text{ in}$ ____ in^2

c. 12 ft, 18 ft; $h = 4 \text{ ft}$ ____ ft^2

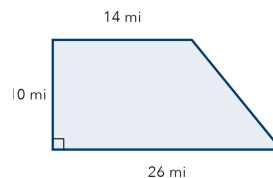
d. 6 m, 10 m; $h = 5 \text{ m}$ ____ m^2

- 2.
- County Land.**
- A county is shaped like this right trapezoid.

a. Sum of the bases ____ mi

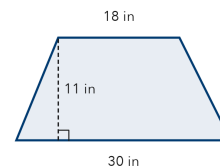
b. Area of the county ____ mi^2

c. Two copies make a rectangle. How long is it? ____ mi



- 3.
- Table Runner.**
- A table runner has the trapezoid shape shown. Find its area.

Show your work.

**Answer** _____ in^2

- 4.
- Sidewalk Panel.**
- A sidewalk panel is a trapezoid with parallel sides 8 m and 20 m and a height of 7 m. Find its area.

Show your work.

Answer _____ m^2

- 5.
- Talia's Answer.**
- A trapezoid has bases 11 cm and 17 cm and a height of 20 cm. Talia says its area is
- 560 cm^2
- . Is Talia right?

Hint Check each step of the formula. Was a step skipped?

Show your work.

Answer _____ cm^2

6. **Pool Cover.** Bases 16 ft and 22 ft, height 10 ft. Fabric costs \$2 per ft². Find the area and cost.

Hint Find the area first. Then multiply by the price.

Show your work.

Area, cost _____

7. **Two Signs.** Sign M has bases 10 cm and 26 cm. Sign N has bases 14 cm and 22 cm. Both are 8 cm tall. Find each area. What do you notice?

Hint Compare the sums of the bases.

Show your work.

Notice _____

8. **Missing Base.** A trapezoid has an area of 84 m² and a height of 7 m. One base is 10 m.

Hint Double the area, then divide by the height.

a. What is the sum of the two bases? _____ m

b. Find the other base. _____ m

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Sofia's Work.** A trapezoid has bases 7 m and 21 m and a height of 5 m. Sofia wrote: " $A = \frac{1}{2} \times 7 \times 5 = 17.5 \text{ m}^2$." Explain the mistake and find the correct area.

TEST PRACTICE

choose the best answer

10. A trapezoid has bases 5 cm and 11 cm and a height of 6 cm. Its area is –

A 22 cm²

B 48 cm²

C 96 cm²

D 165 cm²

11. In $A = \frac{1}{2} \times (b_1 + b_2) \times h$, the first step is to –

A multiply b_1 by h

B divide by 2

C add the bases

D multiply the bases

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

14 A	144 L	98 R	200 F	196 X	280 M	560 Q	380 U	528 Z	264 O
#2b		#3	#4	#5	#6	#7	#8b		

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____